



Fig. 6: Block diagram of the digital PWM peripheral

To generate a frequency of 50,050 Hz, the required period value is 199.8. The PWM period register is set to a value of '200' 80 per cent of the time, and a value of '199' 20 per cent of the time. The value added to the 8-bit accumulator register is $0.8 \times 256 = 204 = 0xCC$.

The table shows how the algorithm works over several PWM periods. The period register of the PWM module is changed between 199 and 200 over time, so the average period of 199.8 is produced.

With the 8-bit accumulator used in this example, the effective frequency can be adjusted in 10Hz steps around the centre frequency of 50,000 Hz. This resolution provides excellent dimming performance in electronic ballast applications.

Power LEDs

LEDs have been widely used in indicator applications for years. The indicator (or winky-blinky) style LEDs are low-current, low-power devices that are not very suitable for lighting applications. However, recent advances in semiconductor manufacturing, silicon structures and phosphor coatings have made high-power LEDs possible.

Present-day power LEDs have a high efficacy that approaches or exceeds that of other efficient light sources. LEDs also offer other benefits like long lifetime and resistance to shock and vibration. These benefits make the LEDs useful for traffic signs, automotive lighting, military applications or any place where safety, reli-

ability or the cost of maintenance could be an issue.

Power LEDs are manufactured with a silicon-carbide or sapphire substrate. The sapphire substrate offers a lower manufacturing cost, but higher thermal resistance. Silicon carbide can be more attractive because of its lower thermal resistance, which is extremely important for power LED applications.

The substrate is then doped with AlInGaP to make the LED red, orange or yellow in colour. Or, the substrate can be doped with AlInGaP to make green, blue or white LED. The colour produced by the LED is a function of the doping (the lens over the LED) and the phosphorous coating that is applied beneath the lens.

White light

The ability to make a white light source with an LED is very important for lighting applications. There are two common methods for making a white LED. The first is to use a blue LED with a phosphor coating that creates a white light. The other method uses an LED that emits light in the ultra-violet range. A mix of red, green and blue phosphors is then used to turn the UV light into visible white light.

There are some pros and cons of each of these methods. The blue-LED+phosphor

method provides a very efficient light source. On the downside, it is difficult to control the exact colour of the light output because of variations in the blue LED. The UV LED+RGB phosphor construction provides a more predictable colour because the properties of the phosphor determine the colour of the light output. A disadvantage of this technique is that the red phosphor tends to degrade faster than other phosphors, causing a shift towards cool white.

Another way to produce white light with LEDs is to use three emitters for red, green and blue. If the LEDs are driven in the right proportions, you get white light. Similar to the UV+RGB phosphor type of LED, the colour of the three-LED solution tends to drift as each LED ages differently. In critical applications, active sensing and control can be used to correct the three-LED white light source, over time.

Thermal issues with LEDs

As stated before, heat dissipation and thermal resistance are big issues with power LEDs. A power LED does not radiate heat. This means, unlike incandescent bulbs, you will not feel the heat when you get near the power LED. Instead, the heat generated by an LED must be mechanically conducted away from the junction.

Power LEDs could not be viable without the same assembly techniques

Using an Accumulator Register to Create High-Frequency Resolution

PWM period	Accumulator value	Carry out from addition	Period register
1st	00	No	199
2nd	00 + CC = CC	No	199
3rd	CC + CC = 98	Yes	200
4th	98 + CC = 64	Yes	200
5th	64 + CC = 30	Yes	200
6th	30 + CC = FC	No	199
7th	FC + CC = C8	Yes	200
8th	C8 + CC = 94	Yes	200
9th	94 + CC = 60	Yes	200
10th	60 + CC = 2C	Yes	200